



Team II

Development and Implementation of the BaSyx Viewer Plugin Extensions

22.05.2026

AGENDA



OUR TEAM / INTRODUCTION

PRODUCT OVERVIEW

REQUIREMENTS

TESTS

DEMO

DOCUMENTS/PM

OUR TEAM



2831182



Manuel Sposato
Projektleiter
KOKI Group



Manuel Sposato
Manujpg

FEHLT



Mattis Weigold
Testmanager
Hitachi Rail GTS



Skullgirl-(Lady-Bugged)
Skullman-G

6084496



Matti Frey
Technische
Redaktion
Lapp Holding



Matti Frey
Matti2603

3652442



Amon Rizzo
Produktmanager
GMG Color



amon1220

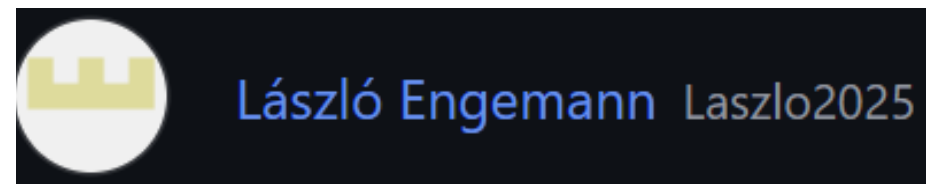
OUR TEAM



5685767



László Engemann
Technische Redaktion
Robert Bosch GmbH



5567934



David Ehrhardt
Systemarchitekt
Hitachi Rail GTS



6490892



Jakob Pauls
Systemarchitekt
Hitachi Rail GTS



Introduction



- Improved the Eclipse BaSyx AAS Web UI
- 2: Implemented sorting and filtering for shells
- 3: Added recursive search for AAS data
- 4: Integrated the Digital Nameplate generator
- 5: Extended metadata support for `createdAt` and `updatedAt`
- 6: Improved boolean labeling in the UI

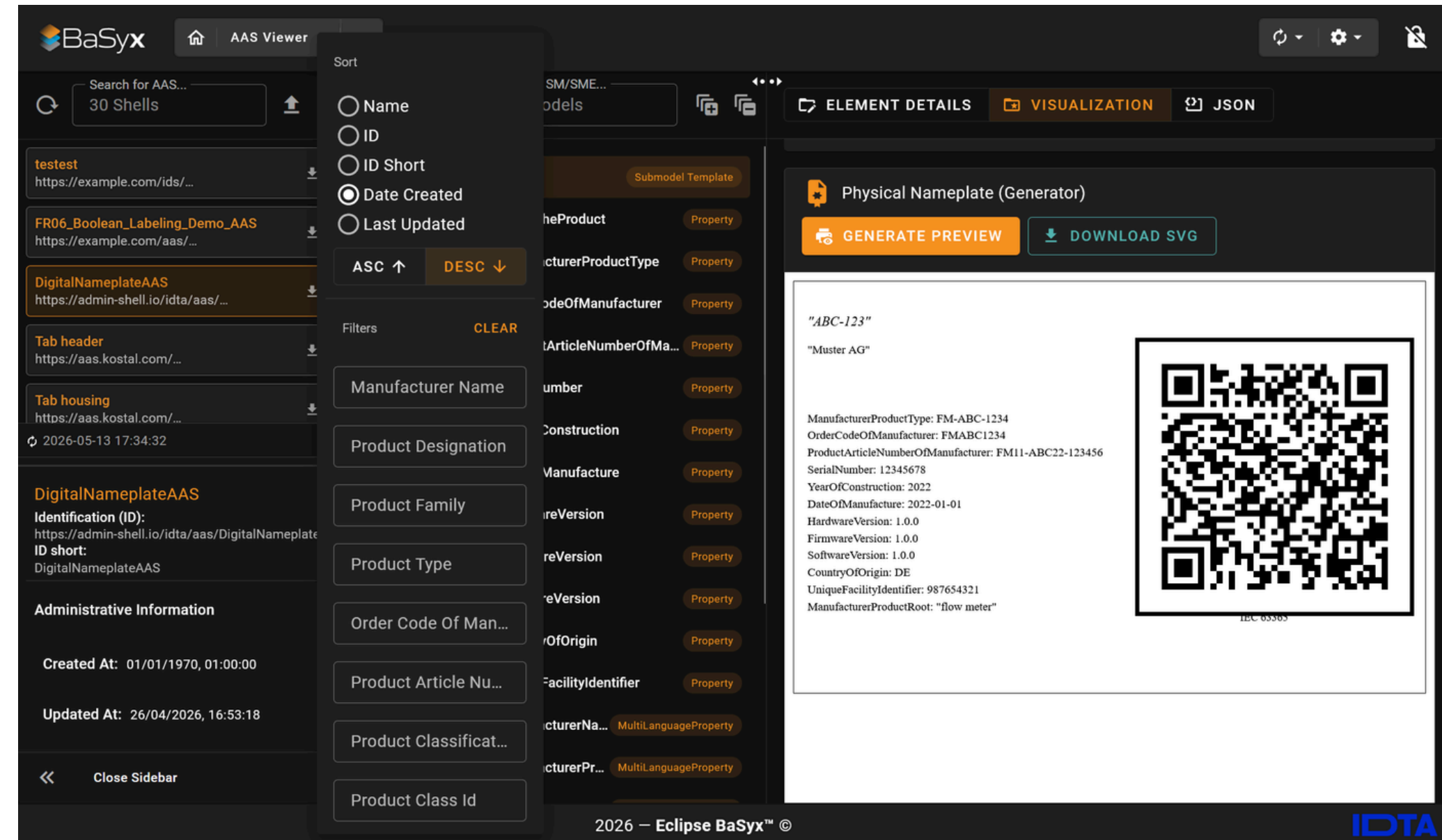


Fig. 1: Overview of BaSyx Viewer

REQUIREMENTS



SORT AAS BY PROPERTIES

- ☒ FR02
- ☐ FR03
- ☐ FR04
- ☐ FR05
- ☐ FR06

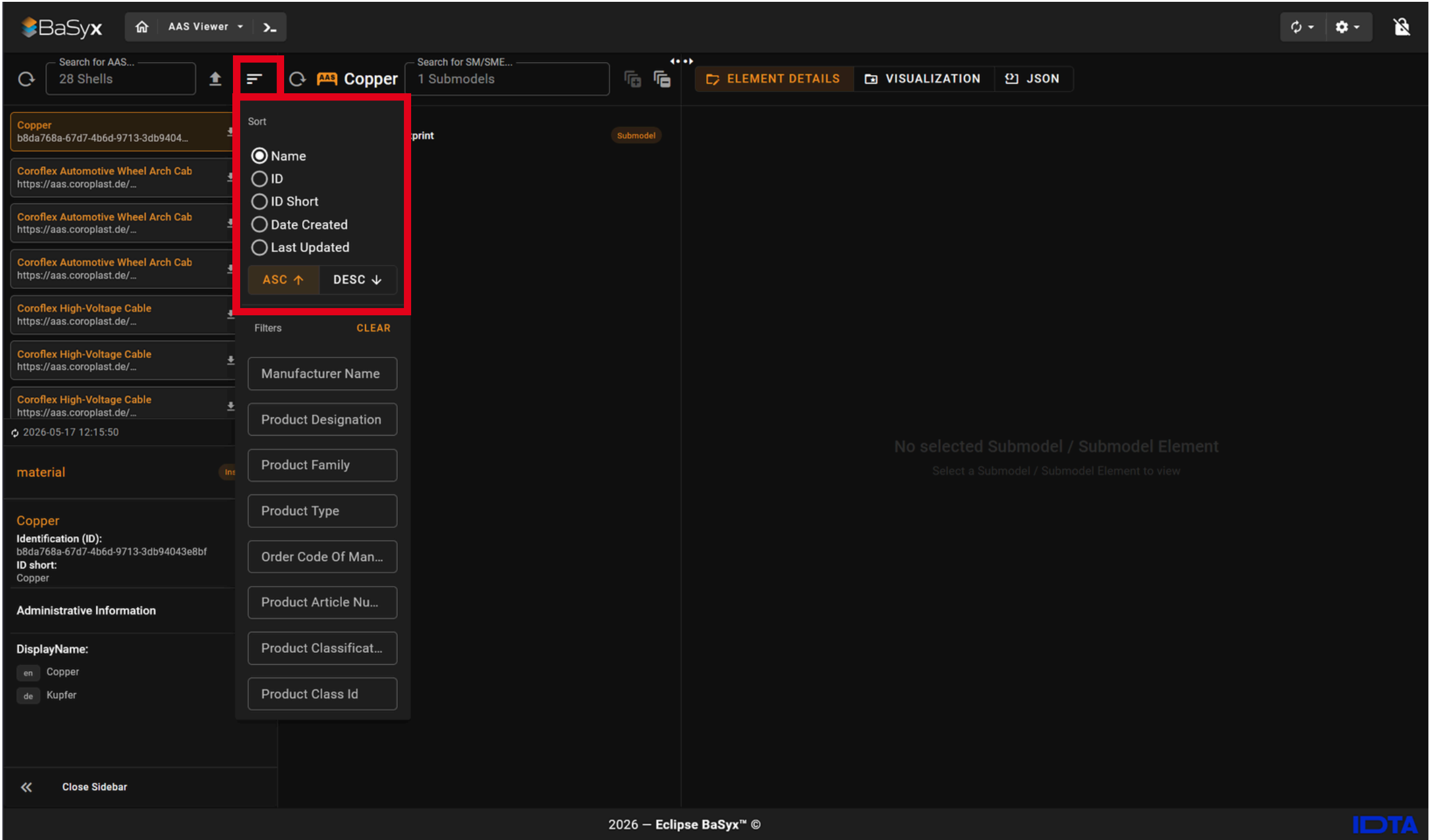


Fig. 2: Overview of BaSyx Viewer Sort

REQUIREMENTS



SORT AAS BY PROPERTIES

SORT MENU

AAS LIST

- ☒ FR02
- ☐ FR03
- ☐ FR04
- ☐ FR05
- ☐ FR06

Search for AAS...		↑	≡
Coroflex Automotive Wheel Arch Cab	https://aas.coroplast.de/...	↓	×
Coroflex Automotive Wheel Arch Cab	https://aas.coroplast.de/...	↓	×
Coroflex Automotive Wheel Arch Cab	https://aas.coroplast.de/...	↓	×
Coroflex High-Voltage Cable	https://aas.coroplast.de/...	↓	×

Sort

- ☒ Name
- ☐ ID
- ☐ ID Short
- ☐ Date Created
- ☐ Last Updated

ASC ↑

DESC ↓

Fig. 4: Excerpt of sorting menu

Fig. 3: List of AAS shells in viewer

REQUIREMENTS



ENHANCED SEARCH FUNCTIONALITY

- FR02
- FR03
- FR04
- FR05
- FR06

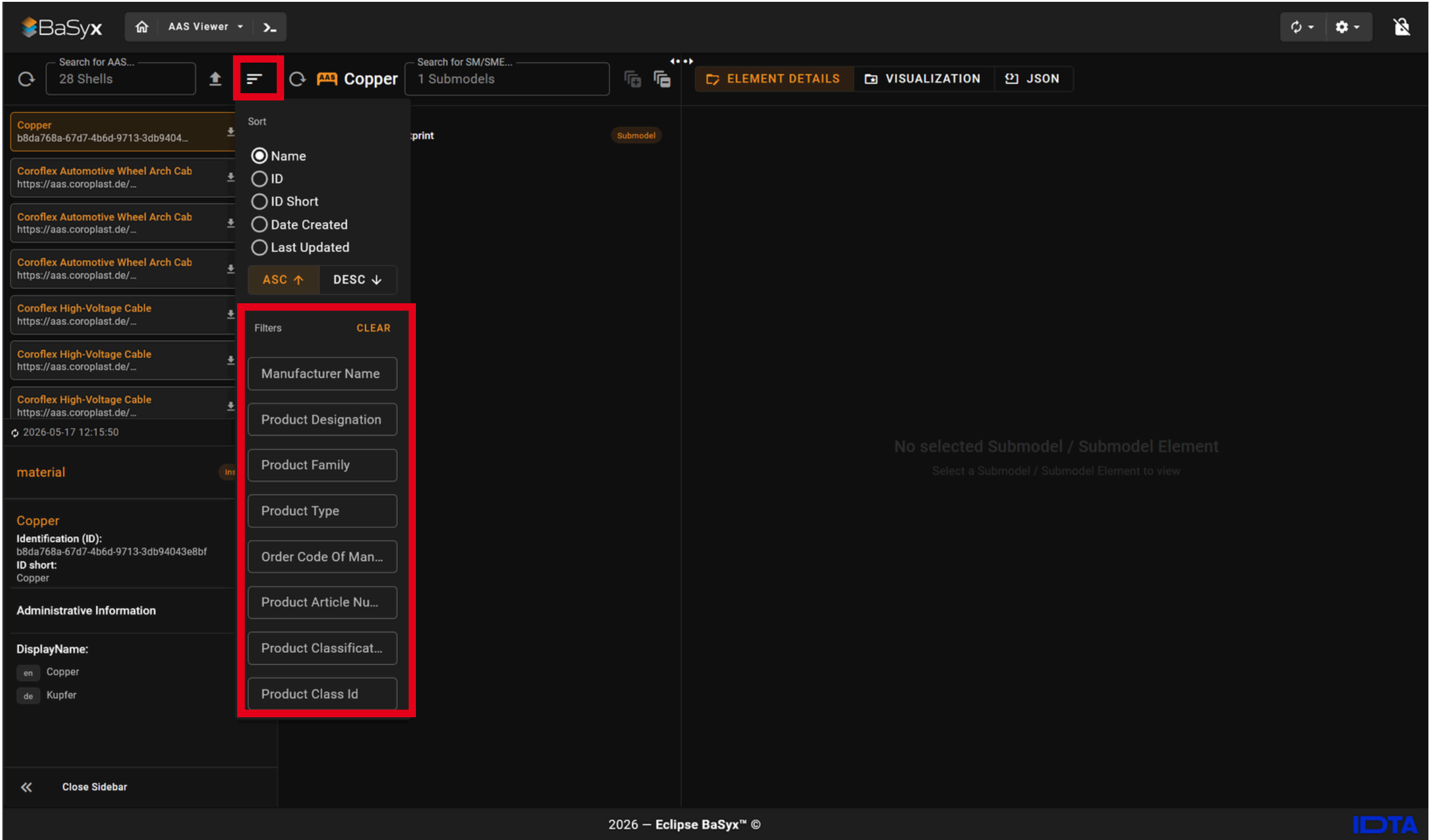


Fig. 5: Overview of BaSyx Viewer Search

REQUIREMENTS



ENHANCED SEARCH FUNCTIONALITY

- ☐ FR02
- ☒ FR03
- ☐ FR04
- ☐ FR05
- ☐ FR06

Definition Layer

aas4j

Logic Layer

basyx-java-server-sdk

uses

Presentation Layer

basyx-aas-web-ui

Rest HTTP calls

REQUIREMENTS



ENHANCED SEARCH FUNCTIONALITY

AAS LIST

- ☐ FR02
- ☒ FR03
- ☐ FR04
- ☐ FR05
- ☐ FR06

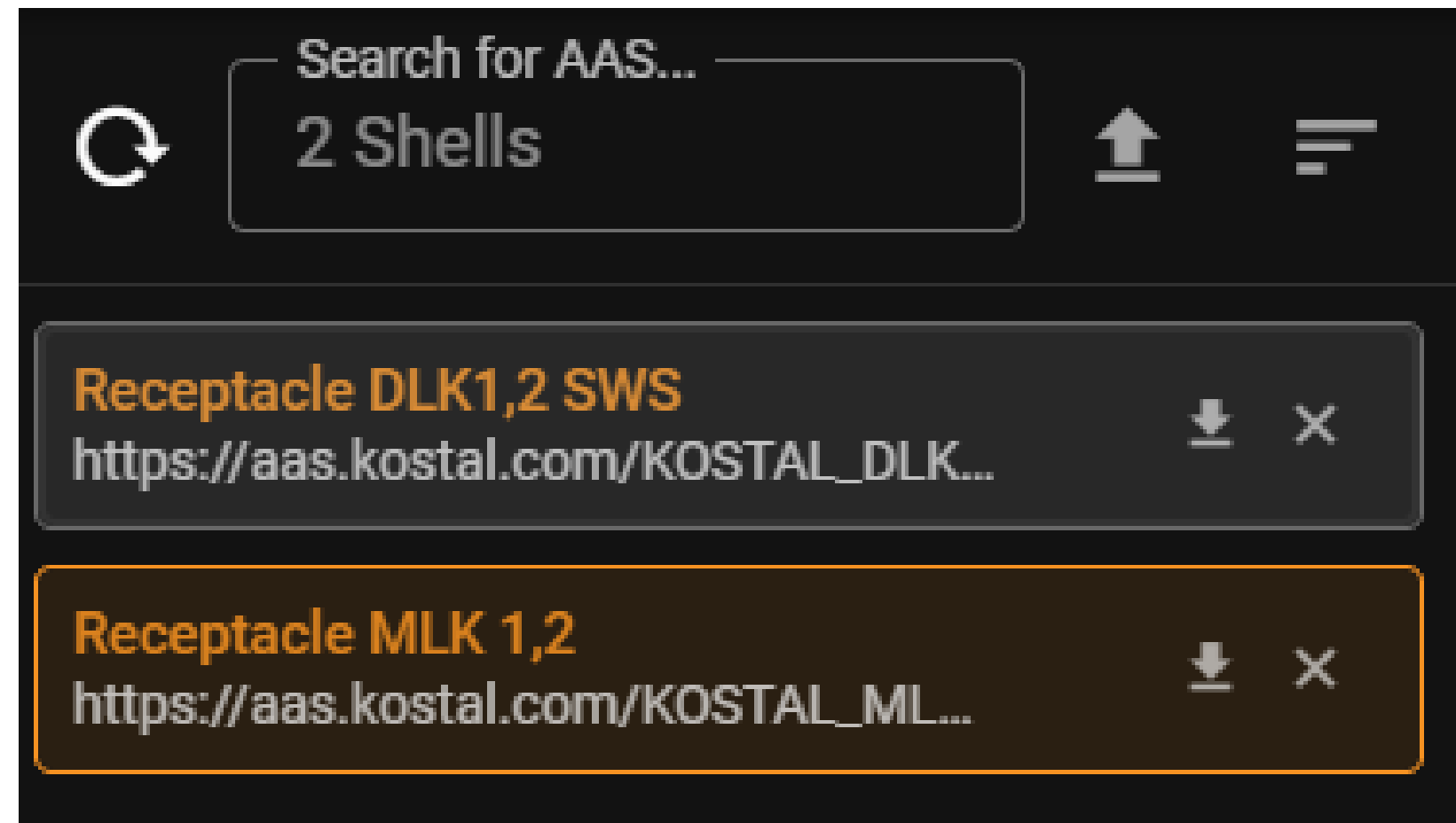


Fig. 6: List of AAS shells in viewer

FILTER MENU

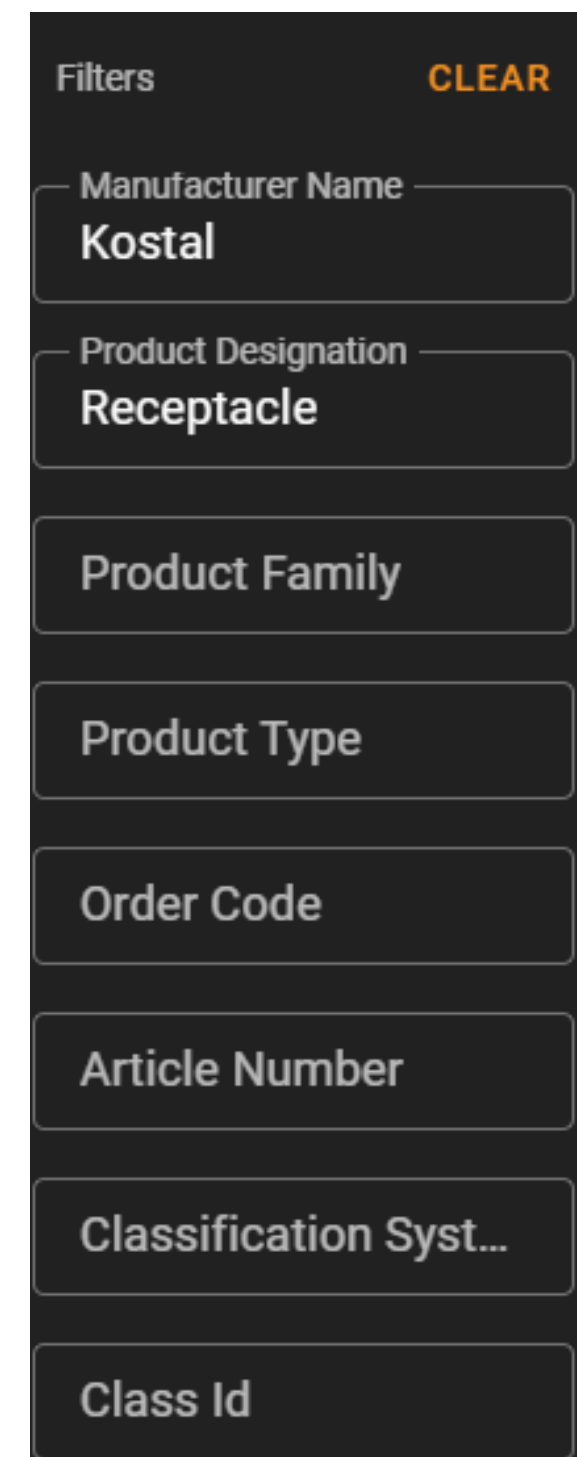


Fig. 7: Excerpt of filtering menu

REQUIREMENTS



INTEGRATE NAMEPLATE GENERATOR

- FR02
- FR03
- FR04
- FR05
- FR06

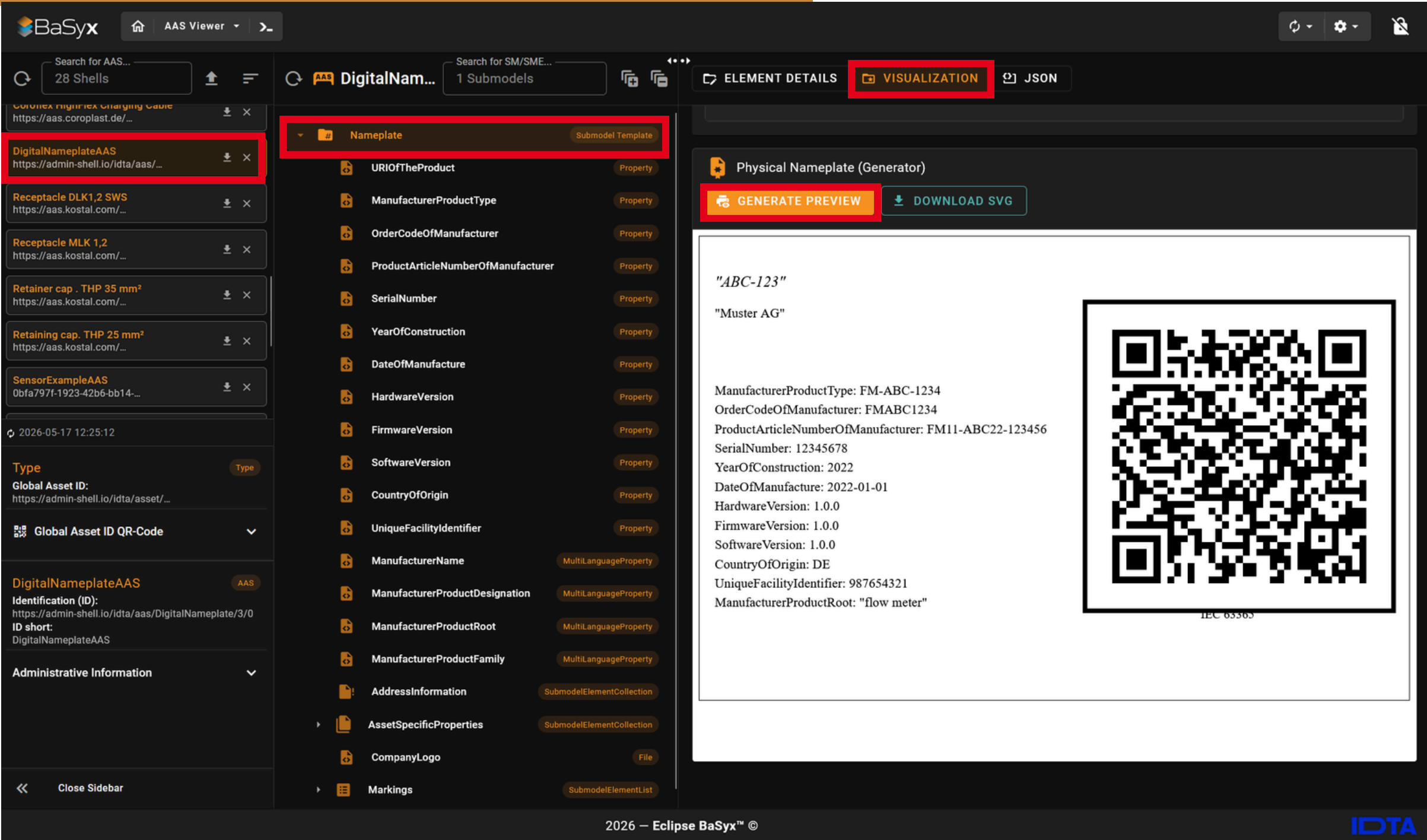


Fig. 8: Overview of BaSyx Viewer Nameplate

REQUIREMENTS



INTEGRATE NAMEPLATE GENERATOR

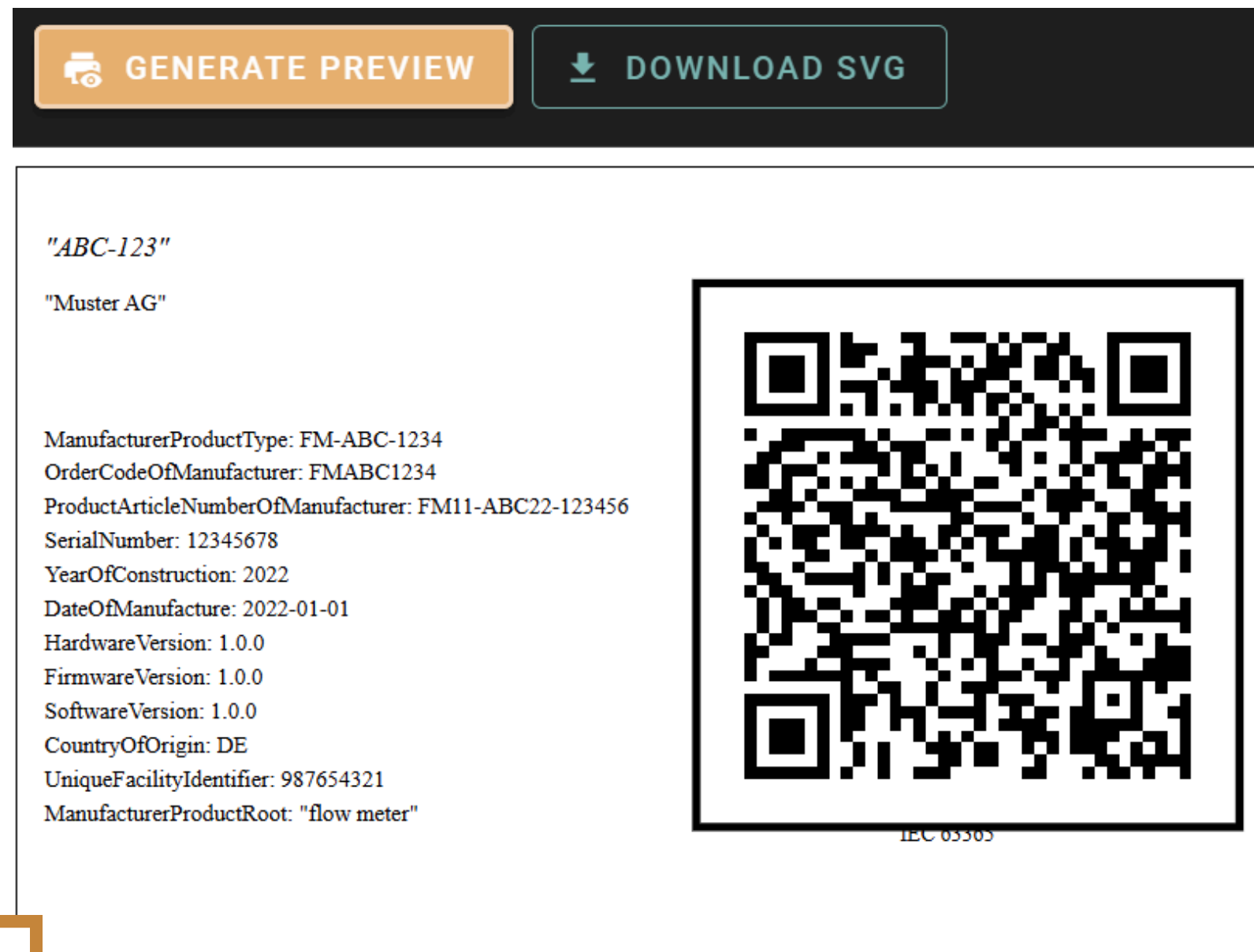


Fig. 9: Sample nameplate

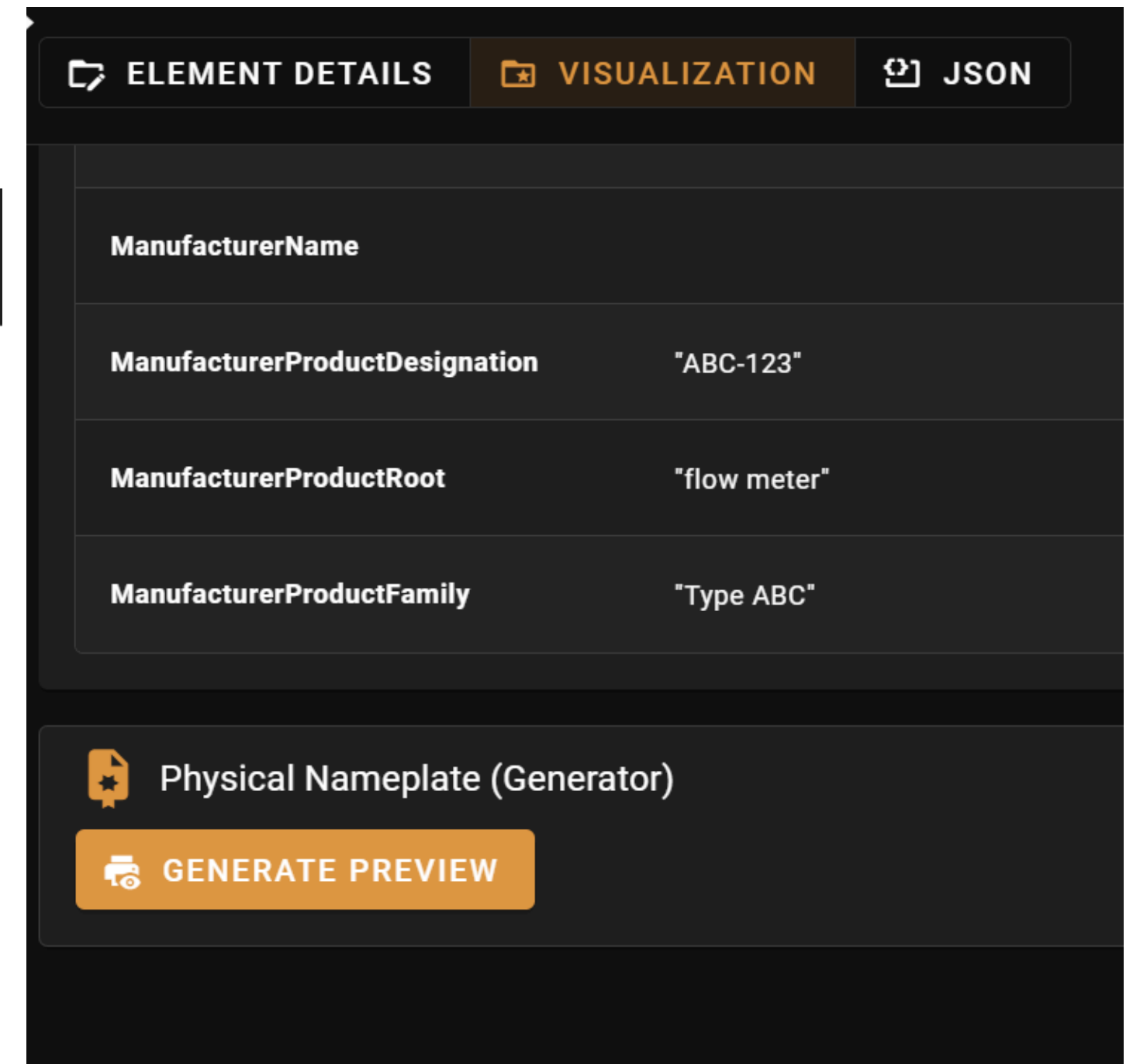


Fig. 10: User interaction for nameplate generator

REQUIREMENTS



INTEGRATE NAMEPLATE GENERATOR

NAMEPLATE3_0.VUE

FRONTEND INTEGRATION

API.JS

LOCALHOST:8080

AAS-BACKEND/NAMEPLATE-GENERATOR-BACKEND

MODIFIED CODE BY OTHER STUDENTS

Fig. 11: Implementation of nameplate generator

☐ FR02

☐ FR03

☒ FR04

☐ FR05

☐ FR06

REQUIREMENTS



API ENHANCEMENTS

- FR02
- FR03
- FR04
- FR05
- FR06

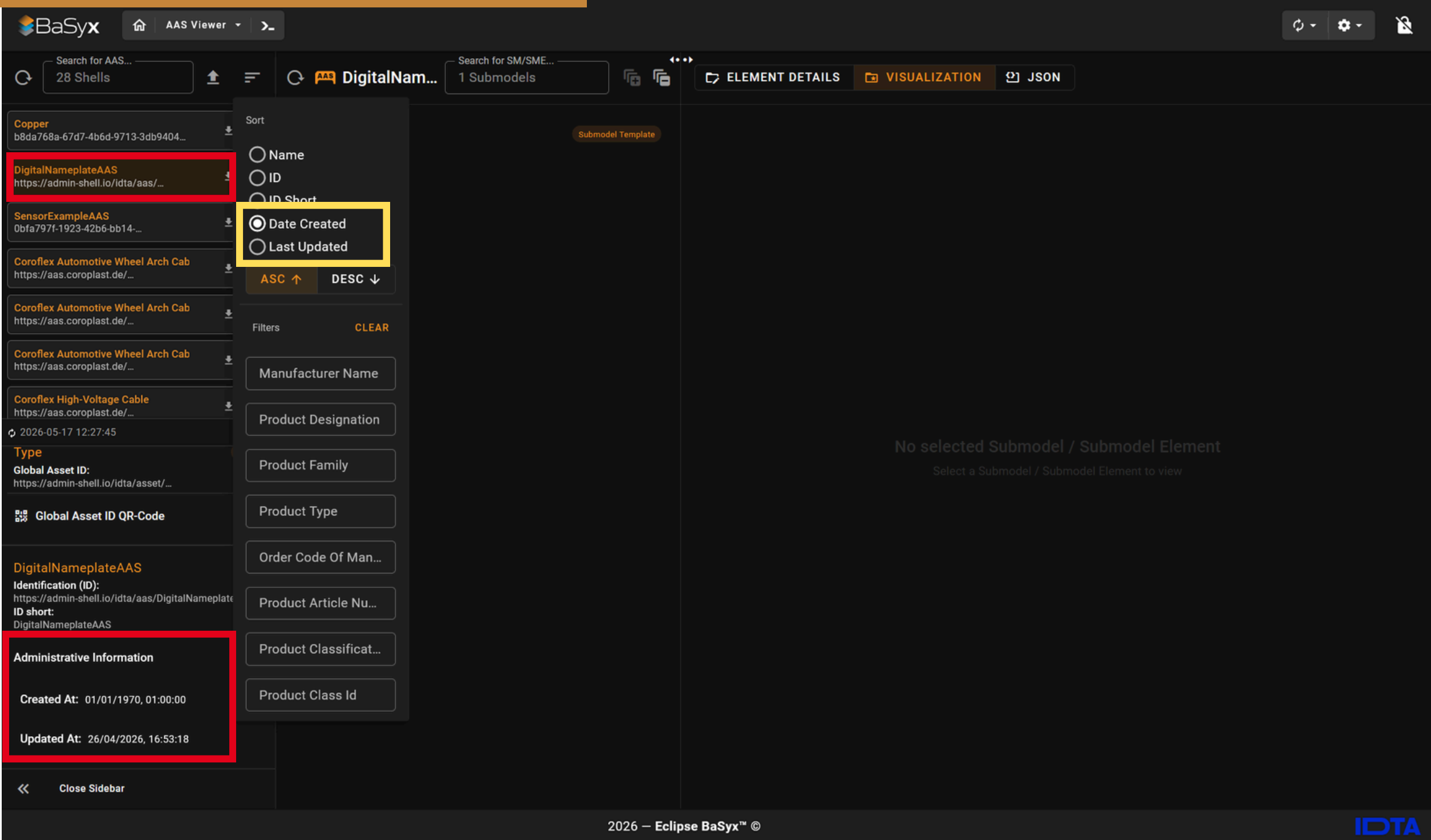


Fig. 12: Overview of BaSyx Viewer created/updated-At

REQUIREMENTS



API ENHANCEMENTS

FRONTEND INTEGRATION

Administrative Information

Created At: 01/01/1970, 01:00:00

Updated At: 02/05/2026, 15:29:32

Fig. 13: Frontend view of cA + uA params

GIVEN SPECIFICATION

Ⓒ AdministrativeInformation *HasDataSpecification*

```
+version: VersionType[0..1]
+revision:
RevisionType[0..1]
+creator: Reference[0..1]
+createdAt: dateTime[0..1]
+updatedAt: dateTime[0..1]
+templateId: Identifier[0..1]
```

Fig. 15: excerpt of IDTA aas-specifications

API RESPONSE

```
modelType: "AssetAdministrationShell"
▼ assetInformation:
  assetKind: "Instance"
  assetType: "SensorType"
▼ administration:
  version: "1.0"
  revision: "0"
  ► creator: { type: "ExternalReference",
  createdAt: "2024-03-22T09:30:00"
  updatedAt: "2024-03-22T09:30:00"
  templateId: "SensorTemplate"
  id: "aas-with-administration-001"
  idShort: "SensorExampleAAS"
```

Fig. 14: Api response from /shells

REQUIREMENTS



IMPROVE BOOLEAN LABELING OF OPERATIONS

- FR02
- FR03
- FR04
- FR05
- FR06

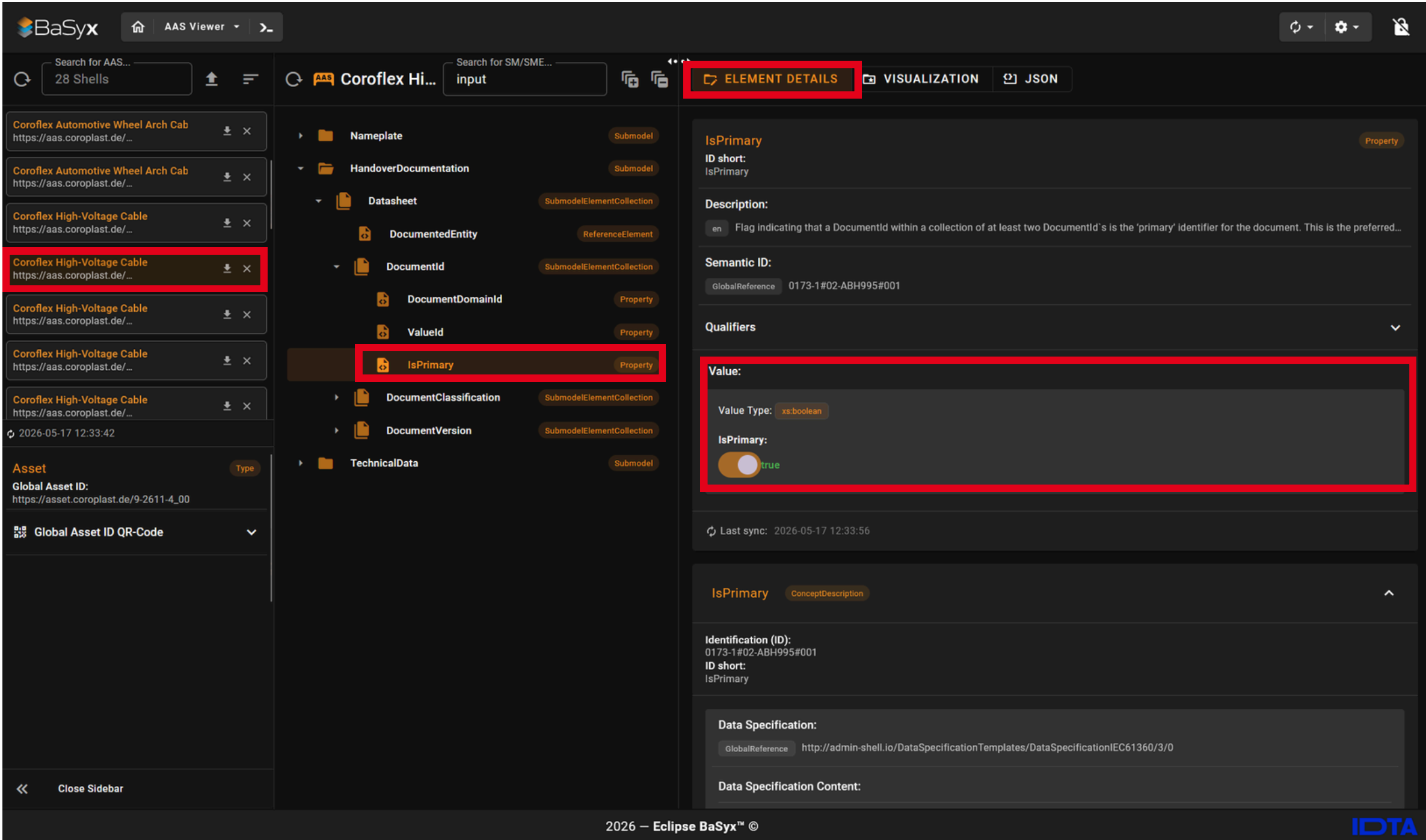


Fig. 16: Overview of BaSyx Viewer Boolean Labeling

REQUIREMENTS



IMPROVE BOOLEAN LABELING OF OPERATIONS

☐ FR02

☐ FR03

☐ FR04

☐ FR05

☒ FR06

Input Variables:

Description:

en Enables or disables the machine....

xs:boolean enableMachine



Fig. 17: boolean labeling true

Description:

en Indicates whether the emergency stop is active....

xs:boolean emergencyStop



Fig. 18: boolean labeling false

TESTS



Req. ID / Name	Functionality	Priority	Testsuit ID or TCS Link
FR.002 / Sorting (SORT)	Dynamic sorting of shells by key attributes.	A	TS.SORT.001.FAS
FR.003 / Filtering (FLTR)	Add Filters for searching shells.	A	TS.FLTR.001.FAS
FR.004 / Nameplate generator integration (GEN)	Integrate submodule into digital nameplate plugin.	B	TS.GEN.001.FAS; TS.GEN.002.FAS
FR.005 / CreatedAt and UpdatedAt support (FIELDS)	Adds two datetime fields to be recieved from the API	D	TC.FIELDS.001.F
FR.006 / Improved labeling (BOOL)	Add extra labels for Boolean values in "Operations" submodules to a visual switch instead of text.	C	TS.BOOL.001.FAS
NFR.001 / Usability	UI improvements must reduce average task time.	--	<i>no testing</i>
NFR.002 / Performance	No significant delay from enhancements.	D	TC.PERF.001.L; TC.PERF.002.L
NFR.003 / Stability	Application should not crash under standard or rapid usage.	--	<i>no testing</i>
NFR.004 / Maintainability	Clear documentation and readable code.	--	<i>no testing</i>
NFR.005 / Licensing	Maintain current open-source license.	--	<i>no testing</i>

Fig. 19: Test plan

DEMO

LIVE-LINK

DOCUMENTS/PM



○ TASK EVOLUTION

○ PROJECT MANAGEMENT

○ GITHUB STATISTICS

1. Original Task

- Search AAS List by idShort, ID and other AAS traits
- Vuetify Combobox and Filter Chips for ID-Short, ID

2. Change during Project

- Search by idShort and ID was already implemented
- Backend version limited efficient filtering for additional attributes

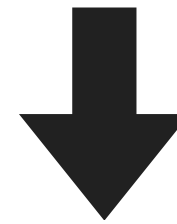
3. Final Scope

- More targeted filters for agreed attributes (Manufacturer Name, Order Code etc.)
- Filtering implemented in the frontend using backend calls

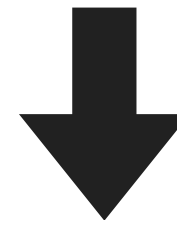
Project process



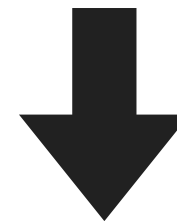
Sep–Nov 2025 Dokumentation & Planung



Mär–Apr 2026 Implementierung



Apr–Mai 2026 Tests



Mai 2026 Rollout & Abschluss

DOCUMENTS/PM

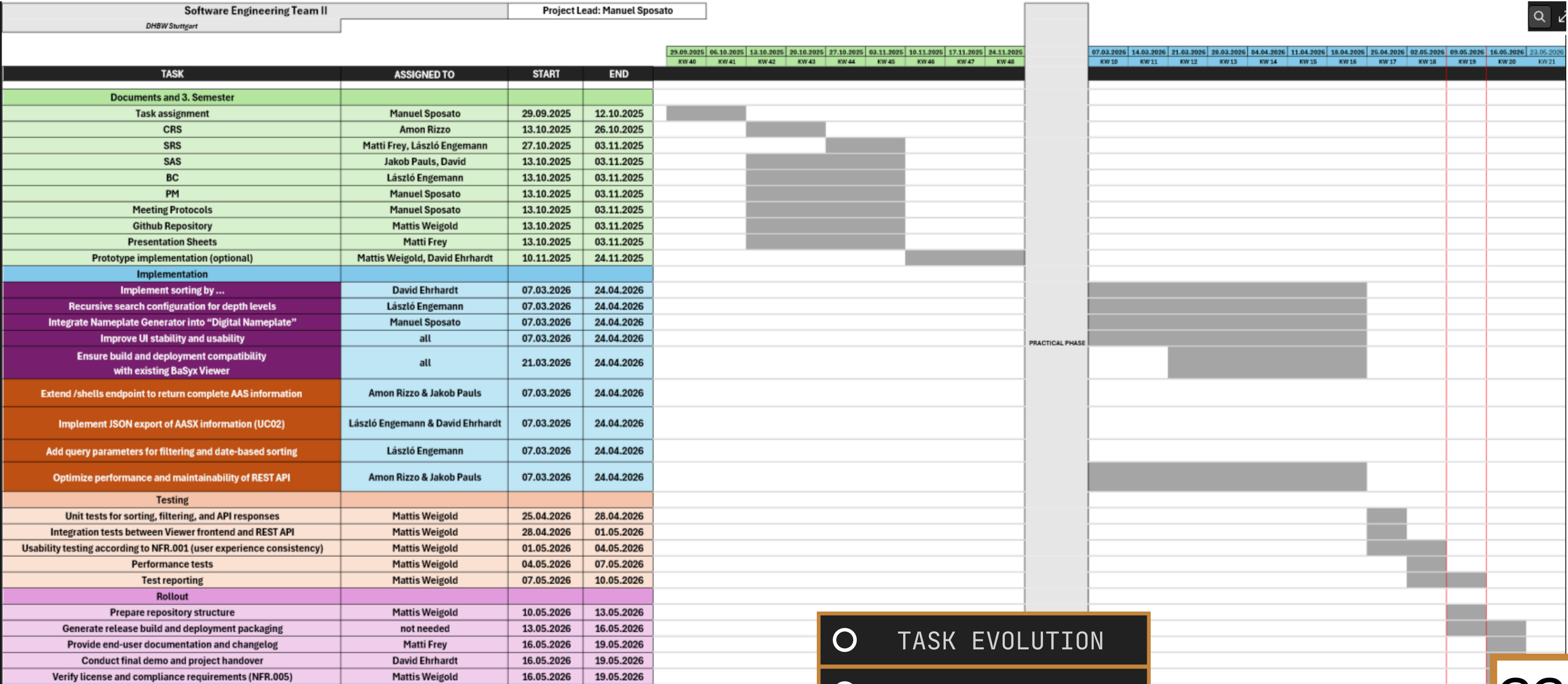


Fig. 20: Gantt diagram

- TASK EVOLUTION
- ◎ PROJECT MANAGEMENT
- GITHUB STATISTICS

DOCUMENTS/PM



- Tasks were organized by project status
- Clear overview of open, ongoing, and completed work
- Feature tasks were assigned to responsible team members
- Development, testing, documentation, and presentation tasks were tracked
- Most feature implementations were already completed
- Remaining work focused mainly on documentation and presentation
- The board helped coordinate responsibilities within the team

- TASK EVOLUTION
- ◎ PROJECT MANAGEMENT
- GITHUB STATISTICS

Fig. 21:
WhatsApp
logo



WHATSAPP + NOTION

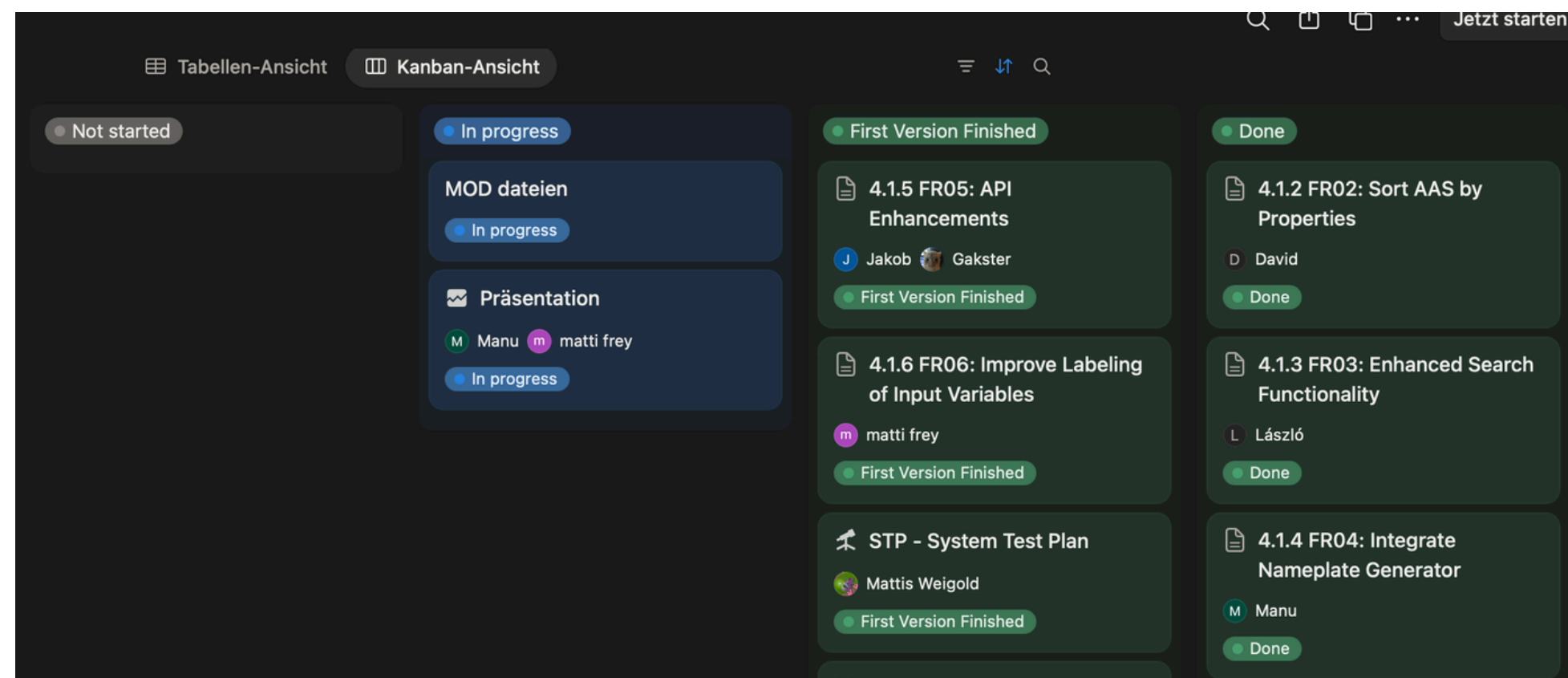
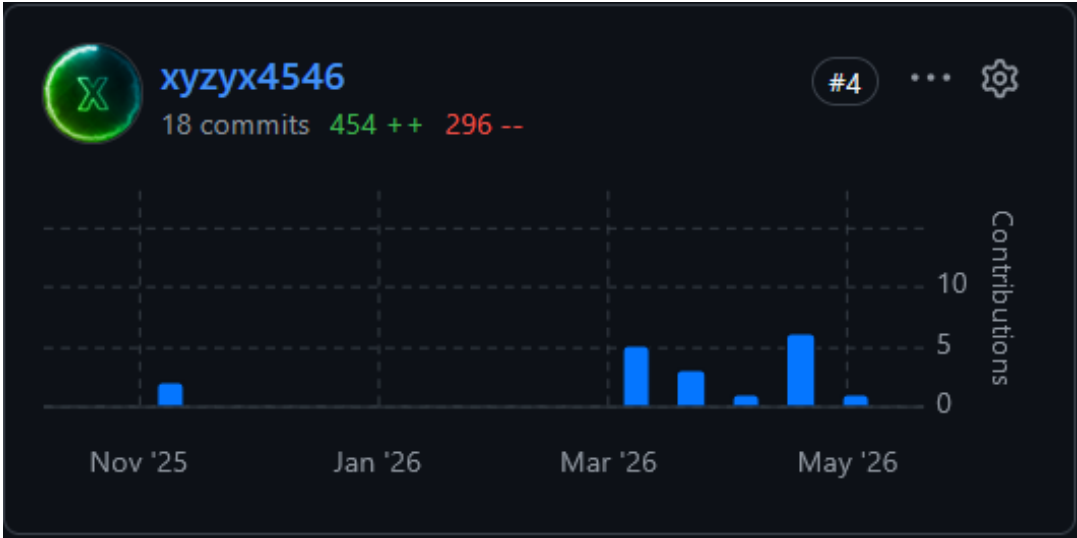


Fig. 22: Kanban board

DOCUMENTS/PM



- ☐ TASK EVOLUTION
- ☐ PROJECT MANAGEMENT
- ☒ GITHUB STATISTICS



Fig. 23: Contributions per GitHub user

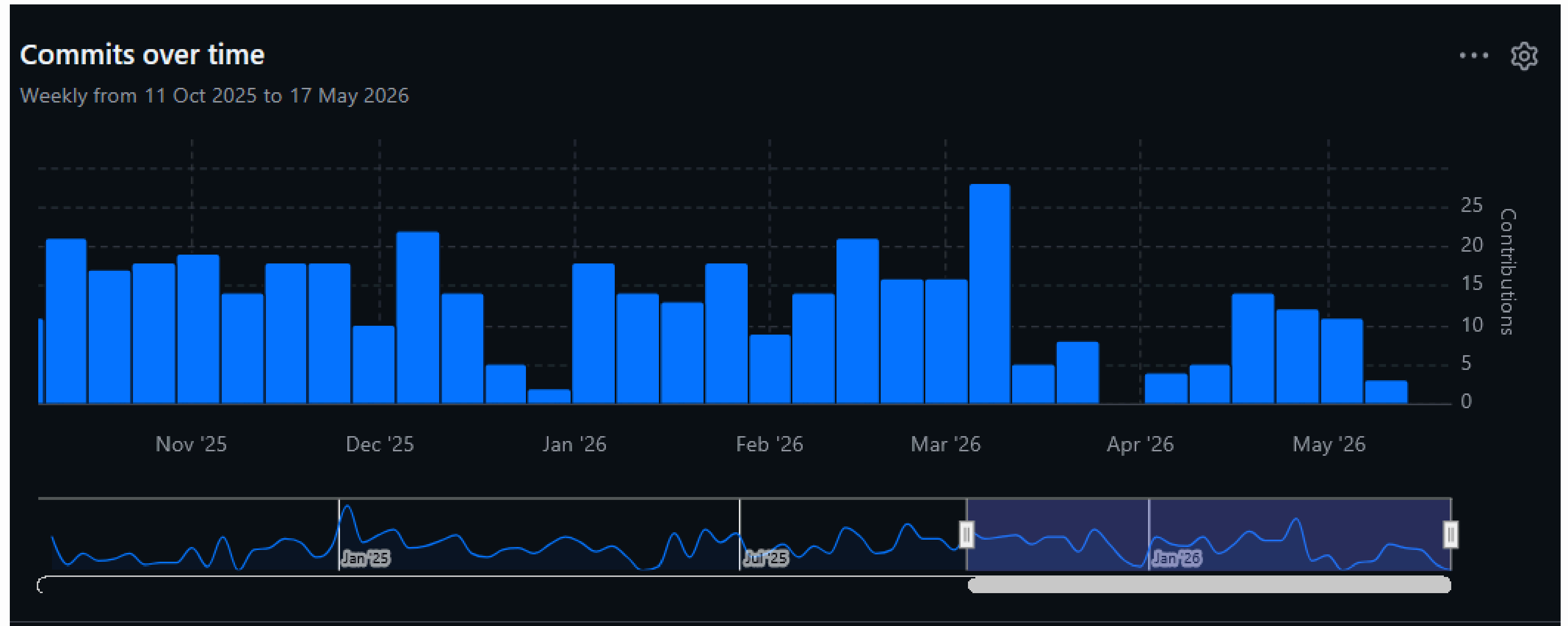


Fig. 24: Commits over time

WHAT COULD BE BETTER

CLEAR REQUIREMENTS

EARLY
SYSTEM
UNDERSTANDING

COMMUNICATION

WHAT WENT WELL

COMMUNICATION

CONTINUOUS
DOCUMENTATION

CLEAR TEAM ROLES

LESSONS LEARNED